

Kyuwon Weon

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EDUCATION

Northwestern University | Evanston, IL

Dec 2026 (expected)

Master of Science in Robotics

Carnegie Mellon University (CMU) | Pittsburgh, PA

May 2022

Bachelor of Science in Mechanical & Biomedical Engineering (University Honors)

SKILLS

Mechanical : PTC Creo, SolidWorks, Onshape, GD&T, DFM, FEA (ANSYS, COMSOL), Rapid Prototyping

Software: Python, C++, C, Linux(Ubuntu), Git, Matlab, Minitab

Robotics: ROS 2, MuJoCo, Gazebo, Rviz, OpenCV, MoveIt, SLAM, PID, Kinematics & Dynamics

EXPERIENCE

Siemens Healthineers | **Manufacturing Intern** | Austin, TX

Jun 2026 - Sep 2026

- Designing and prototyping a multi-actuator packaging automation system from scratch, integrating servo motors, solenoids, and embedded control via Raspberry Pi Pico 2 to fold boxes

Shirley Ryan Ability Lab | **Graduate Researcher** | Chicago, IL

April 2025 - Present

- Implemented a transparent feedforward controller to actively negate back EMF and viscous friction
- Designed manufacturable CAD models for the hip-knee actuation linkages for pediatric exoskeleton

Alcon | **Medical Device Design Engineer** | Fort Worth, TX

Jul 2022 - Jul 2025

- Led the full-lifecycle design of intraocular lens (IOL) implant devices, utilizing Creo and Six Sigma to scale from rapid prototyping to high-volume injection molding
- Executed Root Cause Analysis using DOEs and Minitab statistical methods to resolve critical failure modes
- Collaborated within a multidisciplinary team and moderated clinical prototype evaluations with key opinion leaders (KOLs), translating feedback into design improvements

Alcon | **R&D Intern** | Belmont, CA

Jun 2021 - Aug 2021

- Validated structural deformation predictions of fluid-accommodating IOL models via ANSYS
- Designed custom test fixture in SolidWorks and characterized the dynamic mechanical response of lens haptics under cyclic loading to predict long-term positional stability

PROJECTS

Robot Safety Control Architecture (CBF-QP)

Jan - March 2026

- Architected a 1kHz real-time C++ safety filter, utilizing Control Barrier Functions (CBF) and Quadratic Programming (QP) to mathematically guarantee collision prevention for Franka manipulators in shared space
- Engineered Pinocchio-based kinematics pipeline in C++ with custom ROS 2 interface to handle whole-body constraints and dynamic obstacle interaction

EKF SLAM with TurtleBot3

Jan - March 2026

- Developed a full-stack SLAM pipeline from scratch in C++ with Extended Kalman Filter for TurtleBot3
- Extracted geometric landmarks from raw LiDAR point clouds via custom Euclidean clustering and circle-fitting, utilizing Mahalanobis distance for unknown data association

Jigsaw Puzzle Solver

May 2026

- Developed a vision-guided robotic puzzle solver using OpenCV, Intel RealSense, and an Interbotix PX100 arm for real-time piece identification and pick-and-place assembly

VLM-Driven Interactive Writing Robot

Dec 2025

- Co-developed a VLM-based agent that perceives questions and writes answers using ROS 2 and Python
- Implemented motion planning in MoveIt to generate pen gripping and writing trajectories, while optimizing force/torque safety thresholds to permit writing on a dynamic, handheld whiteboard